

Can Machine Learning Variable Selection and Individual Commodity Fundamentals Improve Futures Return Forecasting?

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Abstract

Contents

1	Introduction	3
1.1	Motivation & Background	3
1.2	The Research Gap	3
1.3	Research Questions & Hypotheses	3
1.4	Contributions to the Literature	3
1.5	Thesis Outline	3
2	Literature Review	4
2.1	Theories of Commodity Futures Returns	4
2.2	Aggregate Commodity-Specific Factors	5
2.3	Individual Commodity Fundamentals	6
2.4	Macroeconomic Predictors	6
2.5	ML Methods in Return Forecasting	6
2.6	Volatility Regimes and Predictability	6
3	Methodology	7
3.1	Baseline Framework (Replication)	7
3.2	Extension I: Penalized Regression	8
3.3	Extension II: Individual Fundamental Predictors	8
3.4	OOS Evaluation Criteria	8
3.5	Portfolio Back-Testing Framework	8
4	Data	9
4.1	Commodity Futures Returns	9
4.2	Macroeconomic Factors	9
4.3	Aggregate Commodity-Specific Factors	9
4.4	Individual Fundamental Predictors	9
4.5	Sample Splits and Descriptive Statistics	9
5	Replication & Baseline Results	10
5.1	In-Sample Estimation (1989-2003)	10
5.2	OOS Predictive Accuracy (2004-2012)	10
5.3	Portfolio Back-Testing (2004-2012)	10
5.4	Replication Assessment	10
6	Extension I: ML Variable Selection	11
6.1	Model Specification and Implementation	11
6.2	Variable Selection Patterns	11
6.3	OOS Forecasting Results - Original Sample (2004-2012)	11
6.4	OOS Forecasting Results - Extended Sample (2013-2024)	11
6.5	Portfolio Back-Testing	11
6.6	Robustness Checks	11

7	Extension II: Individual Commodity Fundamentals	12
7.1	Predictor Construction by Commodity Group	12
7.2	Individual vs. Aggregate Factors: Are They Substitutes?	12
7.3	OOS Forecasting Results (2004-2024)	12
7.4	Combined Model	12
7.5	Portfolio Back-Testing	12
7.6	Regime Interaction	12
8	Synthesis & Discussion	13
8.1	Model Horse Race	13
8.2	Statistical vs. Economic Loss Functions Revisited	13
8.3	Implications for Commodity Market Segmentation	13
8.4	Practical Implications	13
9	Conclusions	14
9.1	Summary of Findings	14
9.2	Limitations	14
9.3	Future Research Directions	14
	References	15
	Appendix	16

1 Introduction

1.1 Motivation & Background

The financialization of commodity markets; why forecasting commodity futures matters to investors, hedgers, and risk managers; the tension between statistical and economic loss functions [1].

1.2 The Research Gap

Three specific limitations of Guidolin and Pedio [1]: fragility of stepwise AIC selection, use of only aggregate commodity factors rather than individual fundamentals, and a sample that ends in 2012 — missing the COVID shock and post-GFC cycles.

1.3 Research Questions & Hypotheses

This thesis aims at answering whether replacing AIC-based stepwise regression with penalized ML methods (LASSO/Elastic Net) improve the out-of-sample forecasting accuracy of commodity futures returns, both statistically and in portfolio returns. Furthermore, it assesses if commodity-specific individual fundamental predictors (inventory levels, crop reports, weather data) add forecasting power beyond aggregate cross sectional factors on an extended 2004-2024 sample.

1.4 Contributions to the Literature

Methodological contribution (ML selection), empirical contribution (individual fundamentals), and out-of-sample robustness contribution (extended sample).

1.5 Thesis Outline

Explain the structure of the thesis.

2 Literature Review

2.1 Theories of Commodity Futures Returns

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Table 1

UMD Factor: Summary Statistics (Jan 1927–Feb 2026)

Statistic	Monthly	Annualized
Mean Return (%)	0.612	7.34
Standard Deviation (%)	4.68	16.21
Sharpe Ratio	-	0.453
t-statistic	4.51	-
p-value (two-tailed)	7.11×10^{-6}	-
Number of Months	1,190	-

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2.2 Aggregate Commodity-Specific Factors

[1]

Guidolin and Pedio [1]

[1]

Guidolin and Pedio

2021

[1, 2]

2.3 Individual Commodity Fundamentals

2.4 Macroeconomic Predictors

2.5 ML Methods in Return Forecasting

2.6 Volatility Regimes and Predictability

As you can see in Table [1](#).

3 Methodology

3.1 Baseline Framework (Replication)

We begin by gathering all T observations of n variables into a data matrix $\mathbf{X} \in \mathbb{R}^{T \times n}$, where each row is a *month* and each column is a macroeconomic variable. Each column is demeaned and standardized to unit variance using the StandardScaler:

$$X_{\text{scaled},ij} = \frac{X_{\text{raw},ij} - \mu_j}{\sigma_j}. \quad (1)$$

Let $\Sigma = V[\mathbf{X}] \in \mathbb{R}^{n \times n}$ denote the variance-covariance matrix of $\mathbf{X}$, which is calculated as:

$$\Sigma = \frac{1}{T-1} \mathbf{X}'\mathbf{X}, \quad (2)$$

and let $\Sigma = \mathbf{U}\mathbf{\Lambda}\mathbf{U}'$ be its spectral decomposition, where $\mathbf{U} \in \mathbb{R}^{n \times n}$ is an orthogonal matrix of order n , whose columns are the eigenvectors $\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_n$ of Σ , and $\mathbf{\Lambda} \in \mathbb{R}^{n \times n}$ is a diagonal matrix of the same order, whose diagonal entries are the corresponding eigenvalues $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n$, which, by convention, are sorted from the largest to the smallest. By multiplying the original scaled data matrix $\mathbf{X}$ with $\mathbf{U}$, we derive the principal components of $\mathbf{X}$ as the elements of the matrix $\mathbf{F} \in \mathbb{R}^{T \times n}$, or formally:

$$\mathbf{F} = \mathbf{X}\mathbf{U}. \quad (3)$$

To assess how much of the total variance in the original n variables is captured by the k -th principal component, or alternatively by the first K principal components, we can compute the variance-covariance matrix of $\mathbf{F}$ as:

$$\begin{aligned} V[\mathbf{F}] &= V[\mathbf{X}\mathbf{U}] \\ &= \mathbf{U}'\Sigma\mathbf{U} \\ &= \mathbf{U}'\mathbf{U}\mathbf{\Lambda}\mathbf{U}'\mathbf{U} \\ &= \mathbf{I}_n\mathbf{\Lambda}\mathbf{I}_n \\ &= \mathbf{\Lambda}, \end{aligned} \quad (4)$$

where we exploited the orthogonality of $\mathbf{U}$. Recalling the fact that $\mathbf{\Lambda}$ is a diagonal matrix, $V[\mathbf{F}] = \mathbf{\Lambda}$ implies that the principal components are mutually uncorrelated, and have a variance equal to the associated eigenvalue. Therefore, the proportion of variance in the original n variables explained by the k -th principal component is given by:

$$\frac{\lambda_k}{\sum_{i=1}^n \lambda_i}, \quad (5)$$

and the cumulative proportion of explained variance by the first K components by:

$$\frac{\sum_{k=1}^K \lambda_k}{\sum_{i=1}^n \lambda_i}. \quad (6)$$

Mirroring the approach used by Guidolin and Pedio [1], we forecast the returns of each commodity j , $R_{t+1,j}$, with the principal component analysis to summarize a large set of macroeconomic factors:

$$R_{t+1,j} = \underbrace{\alpha_j}_{\text{intercept}} + \underbrace{\sum_{i=1}^N \beta_{ij} PC_{t,ij}}_{\text{macroeconomic}} + \underbrace{D_{t,j} \left(\sum_{k=1}^K \gamma_{kj} C_{t,kj} \right)}_{\text{commodity-specific}} + \underbrace{\varepsilon_{t,j}}_{\text{error term}} \quad (7)$$

where α_j is the model's intercept,

3.2 Extension I: Penalized Regression

In the first alternate approach, we keep the principal component analysis and still extract ten principal components from the n initial variables, but instead of running stepwise AIC on those selected principal components, we run a LASSO regression as an alternative way to further reduce the number of parameters to be estimated. With this only being a narrow replacement compared to the original framework, every difference in results should be attributable solely to the selection method.

$$R_{t+1,j} = \alpha_j + \sum_{i=1}^{10} \beta_{ij}^{\text{LASSO}} PC_{t,ij} + \sum_{k=1}^K \gamma_{kj}^{\text{LASSO}} C_{t,kj} + \varepsilon_{t,j} \quad (8)$$

A broader extension to the approach by Guidolin and Pedio [1] is foregoing the principal component analysis entirely and applying LASSO directly to the n variables.

$$R_{t+1,j} = \alpha_j + \sum_{i=1}^{128} \beta_{ij}^{\text{LASSO}} X_{t,i} + \sum_{k=1}^K \gamma_{kj}^{\text{LASSO}} C_{t,kj} + \varepsilon_{t,j} \quad (9)$$

3.3 Extension II: Individual Fundamental Predictors

3.4 OOS Evaluation Criteria

3.5 Portfolio Back-Testing Framework

4 Data

4.1 Commodity Futures Returns

4.2 Macroeconomic Factors

4.3 Aggregate Commodity-Specific Factors

4.4 Individual Fundamental Predictors

4.5 Sample Splits and Descriptive Statistics

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5.1 In-Sample Estimation (1989-2003)

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6.2 Variable Selection Patterns

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6.6 Robustness Checks

7 Extension II: Individual Commodity Fundamentals

7.1 Predictor Construction by Commodity Group

7.2 Individual vs. Aggregate Factors: Are They Substitutes?

7.3 OOS Forecasting Results (2004-2024)

7.4 Combined Model

7.5 Portfolio Back-Testing

7.6 Regime Interaction

8 Synthesis & Discussion

8.1 Model Horse Race

8.2 Statistical vs. Economic Loss Functions Revisited

8.3 Implications for Commodity Market Segmentation

8.4 Practical Implications

9 Conclusions

9.1 Summary of Findings

9.2 Limitations

9.3 Future Research Directions

References

- [1] Massimo Guidolin and Manuela Pedio. “Forecasting Commodity Futures Returns with Stepwise Regressions: Do Commodity-Specific Factors Help?” In: *Annals of Operations Research* 299.1–2 (Apr. 2021), pp. 1317–1356. ISSN: 0254-5330, 1572-9338. DOI: 10.1007/s10479-020-03515-w.
- [2] Ajith S, Manoj Kanti Debnath, and Karthik R. “Statistical and Machine Learning Models for Location-Specific Crop Yield Prediction Using Weather Indices”. In: *International Journal of Biometeorology* 68.12 (Dec. 2024), pp. 2453–2475. ISSN: 0020-7128, 1432-1254. DOI: 10.1007/s00484-024-02763-w.

Appendix

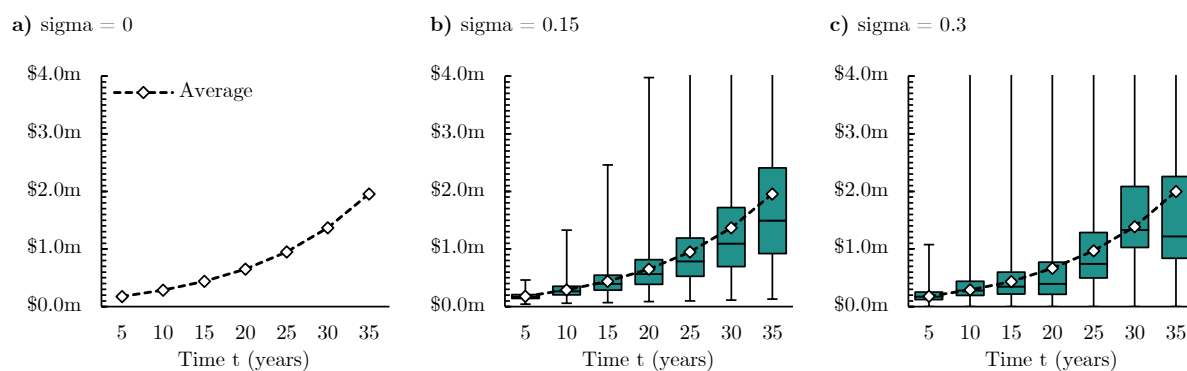
Table 2

Citation Styles with BibLaTeX

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<code>\citeauthor{}</code>	Author	Author
<code>\citeyear{}</code>	Year	Year (not linked)

Figure 1

Example Figure (width should be 16cm)


Code 1

Example Codeblock

```
# KPI OVERVIEW
kpis = pd.DataFrame({
    "Ann. Return (approx)": ann_return_simple,
    "Ann. Return (compounding)": ann_return,
    "Ann. Volatility": ann_vol,
    "Sharpe": sharpe,
    "CAGR": cagr,
    "Max Drawdown": max_dd
})

kpis.T
```